

## RESEARCH ARTICLE - EMPIRICAL

# Prioritizing challenges of agile process in distributed software development environment using analytic hierarchy process

Mohammad Shameem<sup>1</sup>  | Rakesh Ranjan Kumar<sup>1</sup> | Chiranjeev Kumar<sup>1</sup> |  
Bibhas Chandra<sup>2</sup> | Arif Ali Khan<sup>3</sup>

<sup>1</sup>Department of Computer Science and Engineering, IIT (ISM), Dhanbad, Jharkhand, India

<sup>2</sup>Department of Management Studies, IIT (ISM), Dhanbad, Jharkhand, India

<sup>3</sup>College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing, China

## Correspondence

Mohammad Shameem, Department of Computer Science and Engineering, IIT (ISM), Dhanbad, India.

Email: shameem.ism@gmail.com

Handling Editor: Canfora Gerardo

## Abstract

Software organizations are increasingly combining agile methodologies and distributed software development (DSD) for efficiently and effectively built software products. There are numerous challenges associated with scaling agile methods in a distributed environment. Our study is intended to explore and prioritize the challenges for scaling agile practices in the DSD environment. This study was divided into 3 stages. In the first stage, 22 challenges were identified from literature review and grouped into 4 categories: management, team, technology, and process based on discussion with experts. In the second stage, an online questionnaire study was conducted to validate the identified challenges. Finally, analytic hierarchy process method was used to prioritize challenges and their categories based on their relative importance. The results highlighted that management is a most significant category as compared with the other categories. Similarly, lack of management commitments, lack of effective communication, lack of knowledge sharing, etc are identified as the most significant challenges that need to be focused by the organizations for scaling agile methodologies. On the basis of the research findings, we could conclude that the identified challenges along with their categories provide a robust framework to scale agile methodologies in the DSD environment.

## KEYWORDS

agile development, AHP, challenges, distributed software development, multiple-criteria decision making

## 1 | INTRODUCTION

Over the last 2 decades, a variety of models and processes have been developed for the effective management of software development activities. Among these, agile is the most significant process that has been introduced to provide opportunities to manage the dynamic behavior of system development activities.<sup>1,2</sup> Agile is a lightweight software development approach that promotes continuous iterations of development and testing throughout the software development life cycle.<sup>2</sup>

Various agile methods have been developed, and the most popular among these are extreme programming (XP), crystal, scrum, adaptive software development, and feature-driven development.<sup>3</sup> The underlying philosophy of agile methods is derived from the agile manifesto developed by several system development experts who designed different methodologies on the basis of similar ideals.<sup>2</sup> This philosophy primarily focuses on teamwork, the importance of individuals and their interactions, early product delivery, collaboration with the customers, and effective responses to change requirements.<sup>4</sup>

Agile methods were originally designed for small and single-team projects.<sup>2,3</sup> However, few research studies have claimed the successful implementation of agile methods in a distributed environment in large-scale organizations.<sup>3,5</sup> Despite some successful implementations of agile methods,

agile practices are more difficult to scale in a distributed project. Even though agile methods were originally designed for small, single-team projects, which are ideal for agile methods, a distributed project has additional challenges such as communication and coordination issues, cultural differences, and temporal issues.<sup>3,4,6</sup>

Distributed software development (DSD) is a modern paradigm that is used to develop commercially viable software for a global client organization.<sup>7</sup> In a DSD project, the development activities are conducted by virtual teams, in which teams of knowledgeable workers are located in various parts of the world and communicate via different communication channels.<sup>7</sup> Distributed software development offers multiple advantages for client organizations, including low cost, early product delivery, and high-quality products.<sup>8-10</sup> Eventually, because of its economic benefits, various organizations will adopt agile methods to deliver software products that satisfy customer requirements while offering faster development than the traditional development approach.<sup>1</sup>

In spite of the advantages, scaling agile programs in a distributed environment brings additional challenges that increase the complexities in a distributed agile software development environment.<sup>1,11,12</sup> For example, the coordination between teams is considered to be a significant challenge in a distributed agile environment.<sup>11,12</sup> Moreover, distributed projects have additional concerns in terms of interfacing with other organizational units such as marketing, sales, and product management, rendering the development activities more challenging.<sup>7</sup> The challenges in agile development have been widely studied in the available literature.<sup>13-16</sup> In previous studies, most of the challenges have been identified in collocated and large-scale agile projects, rather than specifically addressed for scaling agile methodologies in a DSD environment.<sup>14,15,17</sup> According to Shrivastava and Rathod,<sup>11</sup> agile and DSD processes work on different principles, which is the primary cause of the difficulty in managing software development.<sup>1</sup> Despite these challenges specifically related to the distributed agile process, an increasing trend of applying agile processes in DSD environments in the software industry has emerged.<sup>4,6,18</sup>

Despite the importance of agile development in a distributed environment, the literature indicates that scaling agile methodologies in a distributed environment is a major problem in the software industries, as well as academia. Therefore, this study was designed to suggest improvements in the practices for scaling agile methodologies in a distributed environment by identifying and prioritizing the challenges in a distributed agile software development environment.

The challenges identify the most significant areas that need to be addressed by software organizations. According to Niazi et al,<sup>7</sup> a quality product cannot be delivered on the customer's scheduled time when the challenges are not managed properly. Moreover, the identified challenges can also be used for the internal assessment or prediction of the project.<sup>11</sup> Hence, identifying the challenges in a distributed agile development is firm specific, and prioritizing them can help an organization to improve and revise its management strategies and practices for scaling agile methods in a DSD environment before the initiation of a scaling program. Moreover, it can also facilitate the successful scaling of agile programs and improve the relationships between geographically distributed organizations.

In this paper, we provide a detailed overview of the crucial challenges in distributed agile development. However, these challenges are based on past studies and discussions with only a few agile experts. Hence, the challenges identified from the literature review were empirically verified through the responses collected from a large number of agile practitioners and academicians who have been working (or had experience) in distributed agile development.

As previously discussed, there are few reports of the successful completion of agile projects in a distributed environment because of the lack of investigations into the challenges and their priorities. The various challenges include communication difficulties, a lack of trust between the customers and vendors, cultural differences, and poor technological infrastructure between the sites, which may extend the issues in terms of people, technology, and financing. Hence, all of these challenges must be managed; however, eliminating all of them instantly might not be possible. Therefore, a framework that consists of the challenges and their rankings based on their impacts on scaling practices could be imperative before the initiation of scaling agile programs in a distributed environment. This framework will assist the distributed organizations to focus more on the challenges to attain the best possible outcomes because resources are limited in terms of human resources, technological infrastructure, capital, and the project delivery schedule. Therefore, this study addressed the following objectives:

- to explore the challenges for scaling agile practices in a DSD environment, and
- to prioritize the identified challenges based on their severity and derive key managerial insights.

The rest of this paper is organized as follows. Section 2 presents a literature review to identify the challenges in distributed agile development. A brief discussion of the analytic hierarchy process (AHP) is presented in Section 3. In Section 4, the proposed methodology is presented. Our research findings and some discussion relating to them are presented in Section 5. A comparative study of the proposed framework is discussed in Section 6. Threats to the validity of the study are discussed in Section 7. The implications of the study are discussed in Section 8, and finally, Section 9 presents the conclusion and future work of the study.

## 2 | LITERATURE REVIEW

Various studies have been conducted to highlight the challenges in distributed agile development. The detailed discussions on some of these studies include those of Shrivastava and Rathod,<sup>11</sup> Thomas et al,<sup>12</sup> Layman et al,<sup>17</sup> Niazi et al,<sup>7</sup> Hossain et al,<sup>19</sup> Dikert et al,<sup>3</sup> Alzoubi et al,<sup>20</sup> Laanti,<sup>21</sup> Ghafoor et al,<sup>22</sup> Yagüe et al,<sup>23</sup> and Passivara and Lassenius.<sup>24</sup>

Numerous challenges such as the lack of coordination effectiveness, lack of customer involvement, cultural differences, and temporal distances are considered indicative challenges in distributed agile development.<sup>23</sup> These were also cited by Shrivastava and Rathod.<sup>1</sup> Moreover, Sundararajan et al<sup>25</sup> and Ramesh et al<sup>26</sup> identified some additional challenges, including the lack of communication effectiveness, lack of roles and responsibilities among the team members, inappropriate technology selection, lack of customer involvement, large team sizes, lack of effective communication, and lack of trust between the client and vendor organizations, all of which have been experienced by the practitioners in distributed agile development.<sup>25,26</sup>

Layman et al<sup>17</sup> highlighted the importance of essential communication practices in distributed XP projects, which were also supported by many researchers. In addition, barriers that affect the communication practices in a distributed environment can also be considered to be a challenge in distributed agile projects. In their survey study, Niazi et al<sup>7</sup> found that the distributed dimensions (geographical, cultural, and temporal differences in the DSD) posed significant challenges that created communication difficulties between project stakeholders. According to Alzoubi et al,<sup>20</sup> the challenges reported in distributed agile development included the lack of face-to-face communication, lengthy communication, lack of task awareness, delay in response, unwillingness to communicate, lack of customer involvement, lack of project transparency, lack of trust between a customer representative and vendor, and cultural difficulties. In addition, Thomas et al<sup>12</sup> highlighted the importance of using communication technologies in distributed agile development. Because of their study, instant messaging has been widely used to replace face-to-face communication in a distributed environment.

Specifically, Hossain et al<sup>19</sup> identified the key challenges in the area of distributed agile development. The outcomes of their study were based on a systematic literature review, consolidated into a framework, and discussed in terms of the various categories of the challenges using scrum; mitigation strategies for each risk were also suggested.

Shrivastava and Rathod<sup>1</sup> developed a comprehensive framework to investigate the risk areas in distributed agile development. They classified the risk areas into 5 categories: the software development life cycle, group awareness, technology setup, and external stakeholders, which were mapped according to Leavitt's model of organizational change. In their extended study, Shrivastava and Rathod<sup>11</sup> developed a risk management framework that comprised the risks in the distributed agile process, as well as their causes and strategies, which could be used in industries for managing those risks.

In agile development, the involvement of the customer enables a more efficient development of activities such as writing user stories, discussing product features, prioritizing the feature list, and rapid feedback to the development team.<sup>27,28</sup> Various researchers have reported the role of customer involvement in agile development to be a significant factor for scaling agile methods in a distributed environment. For example, Hoda et al<sup>27</sup> highlighted the importance of customer involvement and its impact on the success of agile development in a distributed environment. They used grounded theory analysis comprising 30 agile practitioners and reported that inadequate customer involvement was one of the key challenges in

**TABLE 1** Challenges in distributed agile development

| Category of Challenges | Challenges                                                     | Reference  |
|------------------------|----------------------------------------------------------------|------------|
| Management             | Cultural differences (CH1)                                     | 1,17,19,23 |
|                        | Lack of motivational activities (CH2)                          |            |
|                        | Lack of management commitments (CH3)                           |            |
|                        | Lack of effective communication (CH4)                          |            |
|                        | Temporal differences (CH5)                                     |            |
| Team                   | Lack of knowledge sharing (CH6)                                | 1,20,23,31 |
|                        | Large team size (CH7)                                          |            |
|                        | Lack of effective involvement of customer representative (CH8) |            |
|                        | Inexperienced developers (CH9)                                 |            |
|                        | Improper task allocation (CH10)                                |            |
|                        | Lack of frequent feedback (CH11)                               |            |
|                        | Lack of trust between customers and vendors (CH12)             |            |
| Technology             | Lack of communication training in DSD (CH13)                   | 17,20,31   |
|                        | Lack of technological resources (CH14)                         |            |
|                        | Budget constraints (CH15)                                      |            |
|                        | Inappropriate selection of communication technologies (CH16)   |            |
| Process                | Poor project scope (CH17)                                      | 4,7,19,29  |
|                        | Lack of project documentation (CH18)                           |            |
|                        | Lack of agile process evaluation mechanism (CH19)              |            |
|                        | Poor requirements and their specifications (CH20)              |            |
|                        | Lack of roles and responsibilities (CH21)                      |            |
|                        | Project size and complexity (CH22)                             |            |

Abbreviation: DSD, distributed software development.

distributed agile projects because they were not included in the development process per the requirement of agile methodologies. Furthermore, Shrivastava and Rathod<sup>1</sup> also highlighted the lack of customer involvement as one of the reasons for defining poor users' stories.

Kasauli et al<sup>29</sup> highlighted project requirement-related challenges in a distributed agile development. The results of their study were based on qualitative approaches, including interviews, and focus group techniques. The challenges in providing effective and efficient development activities in a distributed agile development are related to the core organizational issues such as culture, communication, coordination, and the technological infrastructure between the client and vendor organizations.<sup>1,23</sup> The coordination between the clients and development teams significantly depends on the technological infrastructure between the distributed development sites. Many studies have explored the importance of technological infrastructure for successful agile projects. Herbsleb and Mockus<sup>30</sup> suggested that team members often use communication technologies to communicate and exchange project-related documents among themselves over a network. A poor technological infrastructure can cause other difficulties such as project delays and security issues.<sup>30</sup> In a recent study, Paasivaara and Lassenius<sup>24</sup> conducted a case study for scaling the scrum agile method in a globally distributed organization. In their findings, they reported the need to explore the challenges and management strategies for scaling agile methods in a DSD environment. The challenges for scaling agile methods in a distributed environment identified through a literature review are presented in Table 1.

### 3 | METHOD

#### 3.1 | Analytic hierarchy process

The AHP is the most popular multiple-criteria decision-making technique. It was initially developed by Saaty.<sup>32</sup> Since its introduction, the AHP has been considered by various researchers in a variety of domains for solving complex decision-making problems involving multiple quantitative and qualitative aspects.<sup>33-36</sup> The AHP method is based on the following 3 different phases:

- Decompose a complex decision problem into a hierarchical structure, as shown in Figure 1.
- Determine the priority weight of each factor and subfactor with the help of pairwise comparisons.
- Check the consistency of the judgments.

In the following steps, all 3 phases of the AHP method are briefly discussed.

**Step 1: Decompose a complex decision problem into a hierarchical structure.** In this step, a complex decision-making problem is broken down into a hierarchy of interrelated decision elements.<sup>37</sup> The hierarchical structure of the problem has at least 3 levels, as shown in Figure 1. In the hierarchical structure, the *goal* of the problem is presented at level 1, while the *factor* and the *subfactors* are placed at level 2 and level 3, respectively.

**Step 2: Determine the priority weight of each factor and subfactor with the help of pairwise comparisons:** After decomposing the problem into a hierarchical structure, the priority weights are calculated using a pairwise comparison decision matrix<sup>34</sup> in step 2. At each level, the factors are pairwise compared according to their level of influence and based on the specified criteria at the higher level.<sup>37</sup> In the AHP, a 9-point standardized comparison scale (as listed in Table 2) is used to conduct the pairwise comparisons between the factors to determine the importance of one factor compared with another. A *pairwise comparison matrix* is created<sup>34</sup> for each factor and its corresponding subfactor.

Suppose  $C = \{C_j | j = 1, 2, \dots, n\}$ , where  $n$  is an evaluation factor and every element of the evaluation matrix  $A$ , ie,  $a_{ij}(i, j = 1, 2, \dots, n)$ ,

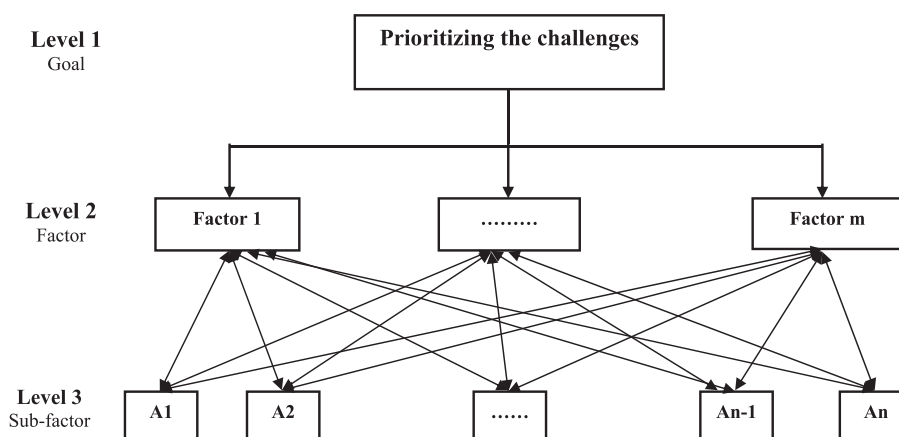


FIGURE 1 Hierarchical structure of problem

**TABLE 2** Description of 9-point scale for intensity of importance

| Definition                   | Intensity of Importance |
|------------------------------|-------------------------|
| Equally important            | 1                       |
| Moderately important         | 3                       |
| Strongly more important      | 5                       |
| Very strongly more important | 7                       |
| Extremely more important     | 9                       |
| Intermediate values          | 2, 4, 6, 8              |

**TABLE 3** Relationship between size of matrix and random consistency index

| Size of the matrix (n)   | 1 | 2 | 3    | 4   | 5    | 6    | 7    | 8    | 9    | 10   |
|--------------------------|---|---|------|-----|------|------|------|------|------|------|
| Random consistency index | 0 | 0 | 0.58 | 0.9 | 1.12 | 1.24 | 1.32 | 1.41 | 1.45 | 1.49 |

represents its normalized relative weight, as illustrated in Equation 1.

$$A = \begin{bmatrix} 1 & a_{12} & \cdots & a_{1n} \\ a_{21} & 1 & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & 1 \end{bmatrix}, \quad (1)$$

where  $a_{ij} = \frac{1}{a_{ji}}$ ,  $a_{ij} > 0$ .

The weight vector  $w$  is determined using the characteristic equation, as shown in Equation 2.

$$Aw = \lambda_{\max} w, \quad (2)$$

where  $A$  is the *pairwise comparison matrix* for the factor,  $w$  is the *weight vector*, and  $\lambda_{\max}$  is the *largest eigenvalue*.

**Step 3: Test the consistency of the pairwise comparison matrix:** In AHP, the pairwise matrix must be consistent.<sup>34</sup> The consistency of the pairwise comparison matrix is determined using the *consistency index* (CI) and *consistency ratio* (CR), with the help of Equations 3 and 4, respectively.

$$CI = \frac{\lambda_{\max} - n}{n - 1}, \quad (3)$$

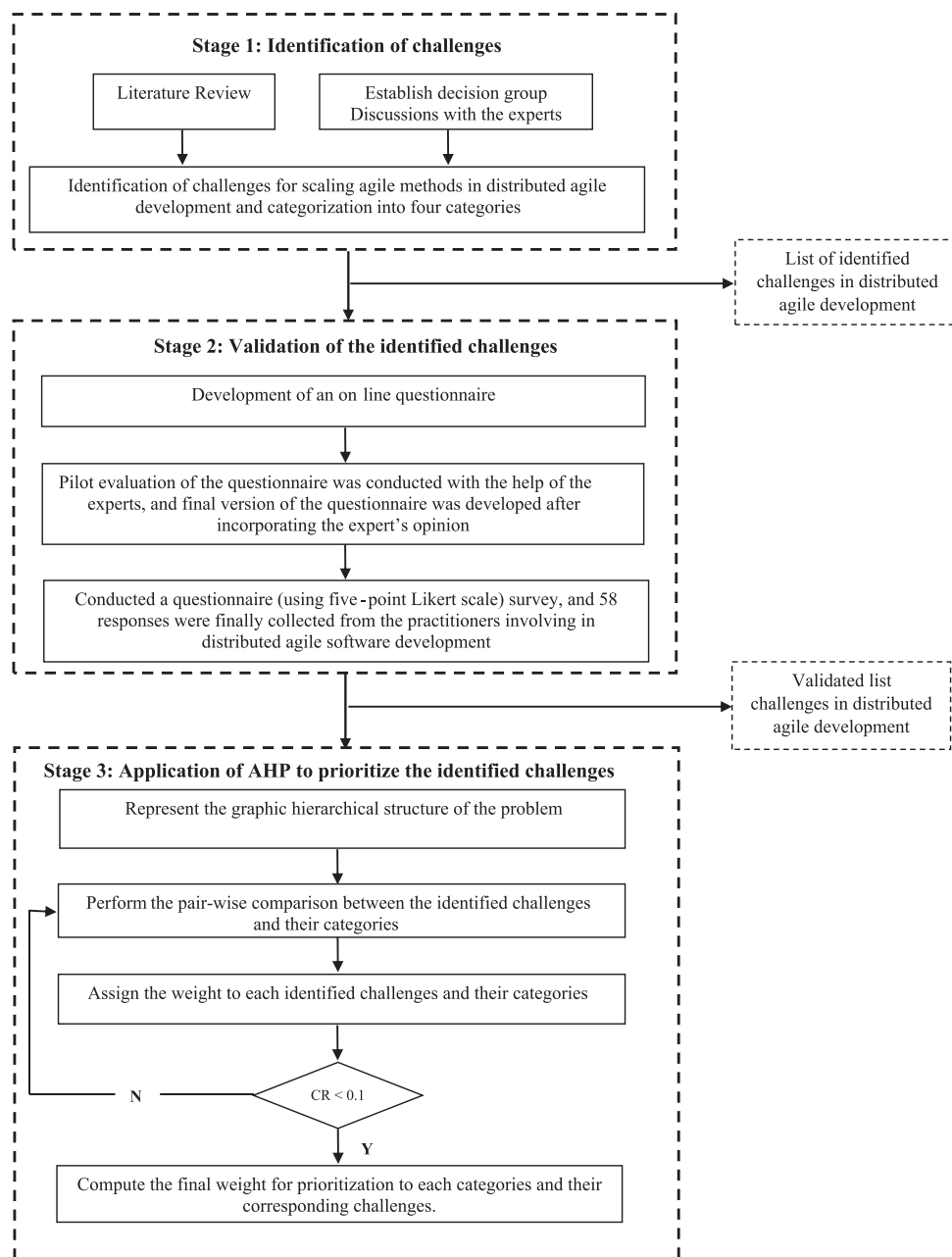
$$CR = \frac{CI}{RI}, \quad (4)$$

where  $\lambda_{\max}$  is the maximum eigenvalue of matrix  $A$  and  $n$  denotes the order of the factors.  $RI$  is the value of a random index of consistency, which has different values based on the number of factors, as listed in Table 3. The accepted value of  $CR$  is up to 0.10. If the calculated value of  $CR$  must be less than 0.10, then the priority vector (weight) of the factor is acceptable and we can conclude that matrix  $A$  has sufficient *consistency*. Otherwise, to improve the consistency, we repeat the evaluation procedure from step 1.

## 4 | PROPOSED METHODOLOGY

The ultimate goal of this study was to identify the challenges and prioritize them for scaling agile methods in a distributed environment on the basis of their importance. The methodology used to achieve these objectives consisted of 3 stages: (1) identifying the challenges in distributed agile development, (2) validating the identified challenges, and (3) using the AHP method to prioritize the challenges. All 3 stages are shown in Figure 2.

First, the challenges in a distributed agile development were identified through a literature review and discussions with 5 experts (2 agile practitioners and 3 academicians). All of the experts selected in our study are directly involved in distributed agile development. They play critical roles in the success of distributed agile projects. The random selection of experts was avoided. Instead, we selected them from the top 3 software organizations, including SAMSUNG, AMAZON India, and Tata Consultancy Services, and from universities such as IIT (ISM), Dhanbad, and City University of Hong Kong. The 22 identified challenges that could affect agile practices were identified and categorized into 4 related to the management, team, technology, and process. Next, the challenges identified from the literature were validated by the questionnaire responses collected from a large number of practitioners and researchers with significant experience in the agile and DSD environment. Finally, the AHP method was used to



**FIGURE 2** Schematic diagram of proposed research design. AHP, analytic hierarchy process; CR, consistency ratio

prioritize the challenges after conducting a pairwise comparison of the challenges. The application of the proposed methodology to validate and prioritize the challenges for scaling agile methods is briefly discussed in the following sections.

#### 4.1 | Questionnaire development

On the basis of the findings of the literature review, an online survey questionnaire to identify the challenges in distributed agile development was developed. According to Khan et al,<sup>38</sup> a questionnaire-based empirical study is a suitable method for collecting self-reporting qualitative and quantitative data from experts using techniques such as questionnaires and interviews.<sup>39</sup> In the present study, the questionnaire was divided into 5 sections. The first section included questions related to the demographics of the respondents, while the other 4 sections consisted of close-ended questions related to the challenges of the management, team, technology, and process categories, respectively. Finally, the questionnaire contained an additional open-ended question to enable the respondent to identify additional challenges that were not mentioned in the questionnaire. The close-ended questions were based on the 22 challenges identified from the literature review and discussions with the experts. We used a 5-point Likert scale, and the possible responses were collected as “strongly disagree,” “disagree,” “neutral,” “agree,” and “strongly agree” in a distributed agile development. According to Niazi et al,<sup>7</sup> adding “neutral” to a 5-point Likert scale does not present any significant disadvantages. Modern researchers believe that the addition of neutral is beneficial because feeling neutral about any statement or topic is a correct response.<sup>40</sup> However, not providing a neutral option will force the respondents to answer positively or negatively, which would be a biased response.

## 4.2 | Pilot testing of questionnaire

To ensure the reliability of the questionnaire, a pilot evaluation was important to assess the validity of its content by measuring the significant variables and their relative importance. In this study, the questionnaire pretesting was conducted with the experts. A final version of the questionnaire was made after incorporating minor corrections based on the review feedback from the experts. During the request for response, the participants were assured that the collected raw data would only be used for research purposes and would not be shared with third parties at any cost. A sample of the survey questionnaire is given in Appendix A.

## 4.3 | Data sources

The aim of this study was to investigate and validate the challenges that influence the scaling programs for agile methods in a distributed context. Hence, it was important to collect data from a diverse range of experts of distributed agile development. On the basis of our experience, and discussions with experts and research colleagues at the IIT (ISM), Dhanbad, we decided to approach the target populations using various online sources, including email, LinkedIn, Facebook, and ResearchGate.<sup>38</sup> There are numerous groups (Agile & Lean development, XP and scrum on LinkedIn, along with software engineering research, agile development, and agile Indonesia on Facebook) related to agile development were joined hosted by LinkedIn (www.linkedin.com) and Facebook (www.facebook.com). These groups provide opportunities to exchange new ideas, knowledge, and information related to growing trends. An online request was posted, and some experts were requested via email to participate in the survey. The online data were collected from July 21, 2017, to October 5, 2017, and the entire data collection process was ended after 1 month and 15 days. At the end of the data collection process, a total of 59 completed responses had been collected. All of the collected responses were manually reviewed by the first author to identify missing or incomplete responses, and one incomplete response was found. After discarding the incomplete response, 58 complete responses were collected for data analysis. The role of the respondents in their organizations ranged from developers to project managers, and all of them had experience with distributed agile development. The demographics of the participants are given in Appendix B.

## 4.4 | Validation of identified challenges

To validate the challenges in distributed agile development, we conducted an online survey of agile practitioners based on the challenges identified in the literature review. The categorization of these challenges is given in Table 4. Table 4 is divided into 3 columns with the categories positive

**TABLE 4** Challenges identified in questionnaire study

| S. No. | Challenges                                               | Analysis of Responses (n = 58) |    |    |          |   |    |         |    |
|--------|----------------------------------------------------------|--------------------------------|----|----|----------|---|----|---------|----|
|        |                                                          | Positive                       |    |    | Negative |   |    | Neutral |    |
|        |                                                          | SA                             | A  | %  | SD       | D | %  | N       | %  |
| CH1    | Cultural differences                                     | 19                             | 27 | 79 | 3        | 5 | 14 | 4       | 7  |
| CH2    | Lack of motivational activities                          | 16                             | 25 | 71 | 4        | 9 | 22 | 4       | 7  |
| CH3    | Lack of management commitments                           | 19                             | 28 | 81 | 4        | 4 | 14 | 3       | 5  |
| CH4    | Lack of effective communication                          | 20                             | 31 | 88 | 2        | 5 | 12 | 0       | 0  |
| CH5    | Temporal differences                                     | 19                             | 28 | 81 | 3        | 4 | 12 | 4       | 7  |
| CH6    | Lack of knowledge sharing                                | 18                             | 32 | 86 | 1        | 4 | 9  | 3       | 5  |
| CH7    | Large team size                                          | 18                             | 28 | 79 | 3        | 5 | 14 | 4       | 7  |
| CH8    | Lack of effective involvement of customer representative | 22                             | 28 | 86 | 1        | 6 | 12 | 1       | 2  |
| CH9    | Inexperienced developers                                 | 18                             | 29 | 81 | 2        | 5 | 12 | 4       | 7  |
| CH10   | Improper task allocation                                 | 16                             | 25 | 71 | 4        | 8 | 21 | 5       | 9  |
| CH11   | Lack of frequent feedback                                | 17                             | 26 | 74 | 3        | 9 | 21 | 3       | 5  |
| CH12   | Lack of trust between clients and vendors                | 20                             | 27 | 81 | 2        | 5 | 12 | 3       | 7  |
| CH13   | Lack of communication training in DSD                    | 17                             | 26 | 74 | 3        | 8 | 19 | 4       | 7  |
| CH14   | Lack of technological resources                          | 21                             | 25 | 79 | 2        | 7 | 16 | 3       | 5  |
| CH15   | Budget constraints                                       | 17                             | 28 | 78 | 3        | 8 | 19 | 2       | 3  |
| CH16   | Inappropriate selection of communication technologies    | 14                             | 26 | 69 | 4        | 9 | 22 | 5       | 9  |
| CH17   | Poor project scope                                       | 18                             | 27 | 78 | 4        | 8 | 21 | 1       | 2  |
| CH18   | Lack of project documentation                            | 16                             | 25 | 71 | 3        | 7 | 17 | 7       | 12 |
| CH19   | Lack of agile process evaluation mechanism               | 15                             | 27 | 72 | 4        | 7 | 19 | 5       | 9  |
| CH20   | Poor requirements and their specifications               | 18                             | 26 | 76 | 3        | 8 | 19 | 3       | 5  |
| CH21   | Lack of roles and responsibilities                       | 19                             | 28 | 81 | 2        | 5 | 12 | 4       | 7  |
| CH22   | Project size and complexity                              | 15                             | 27 | 74 | 3        | 6 | 16 | 6       | 10 |

Abbreviations: A, agree; D, disagree; DSD, distributed software development; N, neutral; SA, strongly agree; SD, strongly disagree.



(strongly agree [SA] and agree [A]), negative (strongly disagree [SD] and disagree [D]), and neutral (N). The positive category in Table 4 indicates the percentage of respondents who agreed with a particular challenge identified in the literature review. The negative category shows the percentage of respondents who did not think that the identified challenge was a significant challenge in distributed agile development. The neutral category represents those respondents who had neutral feelings about the positive or negative categorization of a particular challenge.

The findings demonstrated that the majority of the respondents agreed that the identified challenges could negatively affect the scaling programs of agile methodologies in a DSD environment because all the challenges exhibited a “positive” response of greater than 70 %, except for CH16 (inappropriate selection of communication technologies). Hence, we can conclude that the identified challenges were applicable for further analysis. The method used to validate the questionnaire responses was previously considered in various studies.<sup>7,10,41</sup> In the questionnaire, an open-ended question was presented to enable the identification of additional challenges by the respondents. However, no additional challenges were reported that could have a significant effect on agile development in DSD.

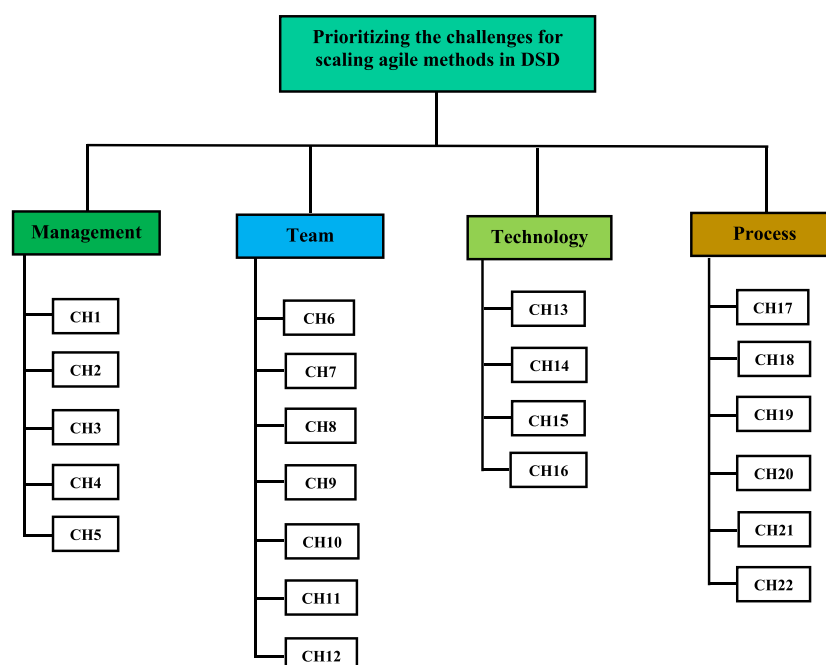
#### 4.5 | Application of AHP to prioritize challenges

The previous sections have identified the challenges and their validation in a distributed agile development. To prioritize them, we used the AHP method, which is briefly discussed in the following steps:

- Step 1.** *Identifying goal, categories (factors), and their corresponding challenges (subfactors).* Table 1 lists the challenges and their corresponding categories.
- Step 2.** *Hierarchical structure of problem.* In this step, we decomposed our problem into a graphical hierarchy structure, as shown in Figure 3. The overall objective of the study is presented at the top level of the hierarchy, while the categories and their respective challenges are shown at level 2 and level 3, respectively.
- Step 3.** *Conducting the pairwise comparison and calculation of priority weight.* We conducted a pairwise comparison of the corresponding challenges within each category, and then the 4 categories were compared and weights were calculated using Algorithm 1. The responses of the experts based on the pairwise comparison of the challenges and their categories are available through the link <https://goo.gl/tt1U5h>. Using the scale values, as listed in Table 2, pairwise comparison matrices were prepared and are presented in Tables 5 and 7 to 9 for the categories management, team, technology, and process, respectively. Similarly, a pairwise comparison between the categories of the challenges is presented in Table 10.

For the pairwise comparison, we synthesized a pairwise comparison matrix by dividing each element of the pairwise comparison matrix by the sum of its column. We obtained the priority vector (weight) of each subfactor by taking the average across the row.

For example, the priority vector of each challenge of the management category is listed in Table 6. The sum of all the priority vectors had to be 1. Here, the priority vector shows the relative weight of a subfactor (challenges in management category). In the management category, CH3 (lack of management commitments) was found to be the most important challenge, followed by CH4, CH1, CH5, and CH2.



**FIGURE 3** Hierarchical structure of the present study for analytic hierarchy process method. DSD, distributed software development



We calculated the degree of consistency (using Algorithm 2) to validate the result. For the pairwise comparison matrix of challenges in the management category, we calculated the largest eigenvalue ( $\lambda_{\max}$ ).

$$\begin{aligned}\lambda_{\max} &= (0.129) * (9.53) + (0.033) * (25) + (0.513) * (1.78) + (0.263) * (4.67) + (0.063) * (16.33) \\ &= 1.22 + 0.825 + 0.9131 + 1.228 + 1.02 \\ &= 5.215.\end{aligned}$$

---

**Algorithm 1** Pairwise comparison

**Require:**  $q_{ij}$ ,  $\triangleright$  where  $i = 1 \dots n, j = 1 \dots m, q_{ij} \in q_{nm}$ . Here,  $q_{ij}$  represents the pairwise comparison value, and  $n$  and  $m$  represent the total number of attributes

**Ensure:**  $P_i$ ,  $\triangleright P_i$  is Eigen vector value for the  $i^{th}$  attribute

```

for (i = 1; i ≤ n; i++) do
  for (j = 1; j ≤ m; j++) do
    For matrix Q, assign the pair-wise comparison values in each  $q_{ij}$ 
  end for
end for
for (j = 1; j ≤ m; j++) do
  for (i = 1; i ≤ n; i++) do
     $q_{ij} = \frac{1}{q_{ji}}$ 
  end for
end for
if (i == j) then
   $q_{ij} = 1$ 
end if
Allocate each value of  $q_{ij}$  into Q,  $q_{ij} \in q_{nm}$ 
// Eigen vector value evaluation
for (j = 1; j ≤ m; j++) do
   $C_j = \sum_{i=1}^n q_{ij}$ 
end for
for (i = 1; i ≤ n; i++) do
  for (j = 1; j ≤ m; j++) do
     $D_{ij} = \frac{q_{ij}}{C_j}$ 
  end for
end for
for (i = 1; i ≤ n; i++) do
   $P_i = \frac{1}{m} \sum_{j=1}^m D_{ij}$ 
end for

```

---



---

**Algorithm 2** Compute consistency

**Require:**  $q_{ij}$ ,  $\triangleright$  where  $i = 1 \dots n, j = 1 \dots m, q_{ij} \in q_{nm}$ . Here,  $q_{ij}$  represents the pairwise comparison value, and  $n$  and  $m$  represent the total number of attributes

**Ensure:** CR,  $\triangleright$  CR represents the value of the consistency ratio

```

for (j = 1; j ≤ m; j++) do
   $S_j = \sum_{i=1}^n q_{ij}$ 
end for
for (j = 1; j ≤ m; j++) do
   $ST_j = S_j \times P_j$ 
  //  $P_j$  represent the eigen vector value, which is calculated by algorithm 1.
   $\lambda_{\max} = \sum_{j=1}^m ST_j$ 
end for
// Compute consistency index (CI) value
 $CI = \frac{\lambda_{\max} - n}{n - 1}$ 
// Compute consistency ratio (CR) value
 $CR = \frac{CI}{RI}$ 

```

---

$\triangleright$  where RI shows the random indices, and CI shows the value of the consistency index

**TABLE 5** Pairwise comparison matrix between challenges of “management” category

| S. No. | CH1   | CH2 | CH3   | CH4   | CH5   |
|--------|-------|-----|-------|-------|-------|
| CH1    | 1     | 5   | (1/5) | (1/3) | 3     |
| CH2    | (1/5) | 1   | (1/9) | (1/7) | (1/3) |
| CH3    | 5     | 9   | 1     | 3     | 7     |
| CH4    | 3     | 7   | (1/3) | 1     | 5     |
| CH5    | (1/3) | 3   | (1/7) | (1/5) | 1     |

**TABLE 6** Synthesized matrix for “management” category

| Management Factor | CH1  | CH2  | CH3  | CH4   | CH5  | Priority Vector (Weight) |
|-------------------|------|------|------|-------|------|--------------------------|
| CH1               | 0.10 | 0.20 | 0.11 | 0.071 | 0.18 | 0.129                    |
| CH2               | 0.02 | 0.04 | 0.06 | 0.03  | 0.02 | 0.033                    |
| CH3               | 0.52 | 0.36 | 0.56 | 0.64  | 0.42 | 0.513                    |
| CH4               | 0.31 | 0.28 | 0.18 | 0.21  | 0.30 | 0.263                    |
| CH5               | 0.03 | 0.12 | 0.08 | 0.04  | 0.06 | 0.063                    |
| $\Sigma=1.000$    |      |      |      |       |      |                          |

Abbreviations: CI, consistency index; CR, consistency ratio; RI, random consistency index.  
 $\lambda_{\max} = 5.215$ , CI = 0.053, RI = 1.12, CR = 0.047 < 0.1 (consistency okay).

**TABLE 7** Pairwise comparison matrix between challenges of “team” category

| S. No.         | CH6   | CH7 | CH8   | CH9   | CH10  | CH11  | CH12  | Priority Vector (Weight) |
|----------------|-------|-----|-------|-------|-------|-------|-------|--------------------------|
| CH6            | 1     | 5   | 3     | 5     | 5     | 1     | 3     | 0.316                    |
| CH7            | (1/2) | 1   | (1/3) | (1/3) | (1/3) | (1/5) | 1     | 0.045                    |
| CH8            | (1/3) | 3   | 1     | 3     | 3     | 1     | 1     | 0.158                    |
| CH9            | (1/5) | 3   | (1/3) | 1     | 1     | (1/3) | (1/3) | 0.067                    |
| CH10           | (1/5) | 3   | (1/3) | 1     | 1     | (1/3) | 1     | 0.078                    |
| CH11           | 1     | 5   | 1     | 3     | 3     | 1     | 3     | 0.234                    |
| CH12           | (1/3) | 1   | 1     | 3     | 1     | (1/3) | 1     | 0.098                    |
| $\Sigma=1.000$ |       |     |       |       |       |       |       |                          |

Abbreviations: CI, consistency index; CR, consistency ratio; RI, random consistency index.  
 $\lambda_{\max} = 7.452$ , CI = 0.0754, RI = 1.32, CR = 0.0559 < 0.1 (consistency okay).

**TABLE 8** Pairwise comparison between challenges of “technology” category

| S. No.         | CH13  | CH14  | CH15 | CH16  | Priority Vector (Weight) |
|----------------|-------|-------|------|-------|--------------------------|
| CH13           | 1     | (1/5) | 3    | (1/3) | 0.118                    |
| CH14           | 5     | 1     | 3    | (1/5) | 0.250                    |
| CH15           | (1/3) | (1/3) | 1    | (1/7) | 0.058                    |
| CH16           | 3     | 5     | 7    | 1     | 0.573                    |
| $\Sigma=1.000$ |       |       |      |       |                          |

Abbreviations: CI, consistency index; CR, consistency ratio; RI, random consistency index.

$\lambda_{\max} = 5.781$ , CI = 0.0521, RI = 0.9, CR = 0.0311 < 0.1 (consistency okay).

We calculated the CI as follows:

$$CI = \frac{\lambda_{\max} - n}{n - 1} = \frac{5.215 - 5}{5 - 1} = \frac{0.215}{4} = 0.053.$$

Here,  $n$  is the number of factors or systems being compared.

We selected an appropriate value for the RI from Table 3, which was found to be RI = 1.12 for the value of  $n = 5$  (number of challenges).

Finally, the CR was calculated as follows:

$$CR = \frac{CI}{RI} = \frac{0.053}{1.12} = 0.047.$$

Here, the value of CR is less than 0.1, and we can say that the priority vector of the challenges is acceptable. Similarly, we find the priority vector (weight) of the challenges for the remaining categories such as team, technology, and process in Tables 7 to 9, respectively. Moreover, in a similar way, the priority vectors (weights) are calculated between the categories of challenges, as given in Table 10.

**Step 4. Ranking of challenges in their corresponding categories (local ranking of challenges).** On the basis of the priority weight given in column 4 of Table 11, the ranks of the challenges in a particular category were computed and are listed in column 5 of Table 11.

**TABLE 9** Pairwise comparison matrix between challenges of “process” category

| S. No. | CH17  | CH18  | CH19 | CH20  | CH21  | CH22  | Priority Vector (Weight) |
|--------|-------|-------|------|-------|-------|-------|--------------------------|
| CH17   | 1     | 2     | 4    | (1/3) | (1/2) | 2     | 0.154                    |
| CH18   | (1/2) | 1     | 2    | (1/6) | (1/3) | (1/2) | 0.072                    |
| CH19   | (1/4) | (1/2) | 1    | (1/5) | (1/4) | (1/2) | 0.051                    |
| CH20   | 3     | 6     | 5    | 1     | 2     | 4     | 0.391                    |
| CH21   | 2     | 3     | 4    | (1/2) | 1     | 3     | 0.236                    |
| CH22   | (1/2) | 2     | 2    | (1/4) | (1/3) | 1     | 0.097                    |

$$\Sigma = 1.000$$

Abbreviations: CI, consistency index; CR, consistency ratio; RI, random consistency index.  
 $\lambda_{\max} = 6.134$ , CI = 0.027, RI = 1.24, CR = 0.022 < 0.1 (consistency okay).

**TABLE 10** Pairwise comparison matrix between “categories” of challenges

| Categories | Management | Team  | Technology | Process | Priority Vector (Weight) |
|------------|------------|-------|------------|---------|--------------------------|
| Management | 1          | 8     | 2          | 4       | 0.533                    |
| Team       | (1/8)      | 1     | (1/4)      | 2       | 0.267                    |
| Technology | (1/2)      | 4     | 1          | 2       | 0.067                    |
| Process    | (1/4)      | (1/2) | (1/2)      | 1       | 0.133                    |

$$\Sigma = 1.000$$

Abbreviations: CI, consistency index; CR, consistency ratio; RI, random consistency index.  
 $\lambda_{\max} = 4.049$ , CI = 0.016, RI = 0.89, CR = 0.018 < 0.1 (consistency okay).

**TABLE 11** Summary of local and global weights of challenges and their rankings

| Category   | Weight of the Categories | Challenges | Local Weight | Local Ranking | Global Weight | Global Ranking |
|------------|--------------------------|------------|--------------|---------------|---------------|----------------|
| Management | 0.533                    | CH1        | 0.129        | 4             | 0.068         | 4              |
|            |                          | CH2        | 0.033        | 5             | 0.017         | 15             |
|            |                          | CH3        | 0.513        | 1             | 0.273         | 1              |
|            |                          | CH4        | 0.261        | 2             | 0.139         | 2              |
|            |                          | CH5        | 0.063        | 3             | 0.033         | 9              |
| Team       | 0.267                    | CH6        | 0.158        | 3             | 0.042         | 3              |
|            |                          | CH7        | 0.067        | 4             | 0.018         | 18             |
|            |                          | CH8        | 0.316        | 1             | 0.084         | 7              |
|            |                          | CH9        | 0.045        | 7             | 0.018         | 14             |
|            |                          | CH10       | 0.234        | 2             | 0.020         | 13             |
| Technology | 0.067                    | CH11       | 0.067        | 6             | 0.062         | 5              |
|            |                          | CH12       | 0.078        | 5             | 0.026         | 11             |
|            |                          | CH13       | 0.118        | 3             | 0.007         | 20             |
|            |                          | CH14       | 0.250        | 2             | 0.016         | 16             |
|            |                          | CH15       | 0.573        | 1             | 0.003         | 22             |
| Process    | 0.133                    | CH16       | 0.058        | 4             | 0.038         | 8              |
|            |                          | CH17       | 0.097        | 4             | 0.012         | 17             |
|            |                          | CH18       | 0.072        | 5             | 0.009         | 19             |
|            |                          | CH19       | 0.051        | 6             | 0.006         | 21             |
|            |                          | CH20       | 0.236        | 2             | 0.031         | 10             |
|            |                          | CH21       | 0.391        | 1             | 0.052         | 6              |
|            |                          | CH22       | 0.154        | 3             | 0.030         | 12             |

**TABLE 12** Prioritization of challenges

| S. No | Challenges                                               | Priority |
|-------|----------------------------------------------------------|----------|
| CH3   | Lack of management commitments                           | 1        |
| CH4   | Lack of effective communication                          | 2        |
| CH6   | Lack of knowledge sharing                                | 3        |
| CH1   | Cultural differences                                     | 4        |
| CH11  | Lack of frequent feedback                                | 5        |
| CH21  | Lack of roles and responsibilities                       | 6        |
| CH8   | Lack of effective involvement of customer representative | 7        |
| CH16  | Inappropriate selection of communication technologies    | 8        |
| CH5   | Temporal differences                                     | 9        |
| CH20  | Poor requirements and their specifications               | 10       |
| CH12  | Lack of trust between clients and vendors                | 11       |
| CH22  | Project size and complexity                              | 12       |
| CH10  | Improper task allocation                                 | 13       |
| CH9   | Inexperienced developers                                 | 14       |
| CH2   | Lack of motivational activities                          | 15       |
| CH14  | Lack of technological resources                          | 16       |
| CH17  | Poor project scope                                       | 17       |
| CH7   | Large team size                                          | 18       |
| CH18  | Lack of project documentation                            | 19       |
| CH13  | Lack of communication training in DSD                    | 20       |
| CH19  | Lack of agile process evaluation mechanism               | 21       |
| CH15  | Budget constraints                                       | 22       |

Abbreviation: DSD, distributed software development.

**Step 5.** *Global weight of challenges (final rank of challenges).* In AHP, the global weight explains the contribution of a particular factor to the overall objective of the study. In the hierarchical structure, the global weight of each challenge is the product of the local weight of a challenge times the weight of its respective category.

For example, the global weight for challenge CH1 is  $0.533 * 0.129 = 0.068$ .

In column 6 of Table 11, we give the global weight for each challenge of the respective category.

**Step 6.** *Prioritizing challenges.* In the final step, the global weight was used to evaluate the final rank of each challenge. Table 12 summarizes the findings. The challenge "CH3: lack of management commitments" is found to be the most important factor, whereas "CH15: budget constraint" is the least important factor among the 22 identified challenges for scaling agile methodologies in a DSD environment.

## 5 | RESEARCH FINDINGS AND DISCUSSIONS

In Table 12, the challenges in a distributed environment for scaling agile methodologies are listed on the basis of their level of importance.

CH3 (lack of management commitments) was found to be the most significant factor playing an important role for scaling agile practices in a distributed context. According to Kontio et al,<sup>42</sup> management commitments can be defined as "the degree to which the higher and lower level management in an organization financially support, realize, and are involved in agile practices." Chow and Cao<sup>43</sup> discussed the importance of management commitment in their empirical study. They reported that when an agile practitioner or manager who is highly committed to implementing agile practices in DSD was replaced by someone who had lower commitment, all of the prior activities were lost.

The AHP results indicated that CH4 (lack of effective communication) was the second most important challenge influencing scaling programs of agile methodologies in a distributed context. Kuusinen et al<sup>44</sup> mentioned that communication is a primary concern in a distributed agile environment and has a direct impact on the success of agile projects. There are generally 2 types of communication, including synchronous and asynchronous communications. In colocated agile development, communication is synchronous, with team members interacting face-to-face with each other to exchange knowledge management. However, the DSD process is different from a colocated process in terms of geographic distance, cultural distance, and time zones. Thus, frequent face-to-face communication is not possible among the team members without traveling to the other development sites.<sup>45</sup> This lack of face-to-face communication can lead to misunderstandings, limited social interaction, delayed knowledge transfer and feedback, and a lack of trust between the team members working on a distributed agile project.<sup>46</sup>

CH6 (lack of knowledge sharing) was reported to be the third most important challenge in a distributed agile project. According to De Brito et al,<sup>15</sup> knowledge sharing is a process in which knowledge is transferred from one person to another and may take place in a planned or natural manner as

a result of another activity. Effective communication is necessary to exchange knowledge or ideas among the team members.<sup>47</sup> A lack of effective communication among the team members is a major reason for a lack of knowledge transfer among them.<sup>24</sup> In a distributed agile project, a poor technological infrastructure can lead to difficulties in knowledge sharing among the distributed agile developers. For instance, knowledge sharing becomes the most critical factor when a new employee joins the project team; a lack of awareness about the project and his responsibilities can lead to additional difficulties for knowledge sharing between the client and vendor organization.<sup>48</sup>

On the basis of the outcomes of AHP, CH1 (cultural differences) was mentioned as the fourth most significant factor affecting scaling programs for agile methods in a distributed environment. It is a fact when the distributed agile development team members or customers operate at distributed sites and have different cultural backgrounds.<sup>49,50</sup> In a distributed agile process, development activities strongly depend on the effective coordination of team members. Hence, it is important to manage the cultural differences for the successful completion of the project.<sup>49</sup> For instance, culturally different people may not effectively communicate using their native languages. Thus, information can be misinterpreted among the stakeholders, which may create misunderstanding and confusion between the client and vendor organizations. Moreover, it has been observed that cultural differences can give rise to other difficulties such as communication issues and a lack of trust between the client and vendor organizations.<sup>7</sup>

CH11 (lack of frequent feedback) was reported as the fifth most challenging factor influencing the successful implementation of agile practices in a distributed environment. Instantaneous feedback/responses among the developers and practitioners is a fundamental principal in agile development. A lack of frequent feedback among the developers can cause delays in the project schedule. Herbsleb and Mockus<sup>30</sup> reported that team members in a DSD environment use communication technologies to communicate among themselves. In addition, project-related documents are shared over a network. A poor technological infrastructure and large time zone differences can cause other difficulties such as project delay and security concerns about sharing important documents between the development sites.<sup>15</sup>

CH21 (lack of roles and responsibilities) was found to be the sixth ranked challenging factor influencing agile methodologies in a distributed environment based on the outcomes of the AHP process. According to Bayona-Oré et al,<sup>51</sup> a lack of clearly defined specific roles and responsibilities assigned to team members and customers may create difficulties in using the agile methods in a distributed context. Therefore, it is important that the roles and responsibilities be clearly defined; otherwise, confusion may occur during the implementation of agile practices in a DSD environment.<sup>51</sup>

On the basis of the findings of the present study, CH8 (lack of effective involvement of customer representative) was mentioned as the seventh important challenge influencing the operation of agile methodologies specifically when executed in a distributed environment. According to Hoda et al,<sup>27</sup> agile methods expand the role of the customers throughout the project life cycle rather than during the initial development phase. This customer involvement makes various activities more efficient like writing user stories, discussing product features, prioritizing the feature list, and providing rapid feedback to the development team.<sup>23,28</sup> The limited overlapping of working hours and a poor technological infrastructure between the development sites results in an increase in the difficulties for customer involvement, which can create difficulties in the successful implementation of user stories into the systems.

## 6 | COMPARISON OF PROPOSED FRAMEWORK

Table 13 compares the identified challenges of the proposed model, those of Shrivastava and Rathod,<sup>1</sup> and the Standish group's chaos report.<sup>52</sup> Shrivastava and Rathod<sup>1</sup> identified 45 risk factors in 5 different categories of project management, group awareness, software development life cycle, stakeholders, and technology and developed a framework for distributed agile development. Our study was similar to that of Shrivastava and Rathod.<sup>1</sup> Therefore, we compared the identified challenges of our proposed framework with those listed by Shrivastava and Rathod for agile development. In a recent survey report conducted by the Standish group,<sup>52</sup> the participants were asked about the factors that caused software projects to be challenging. In their findings, the 8 most challenging aspects were reported for software development activities that negatively impact the success of software projects.

## 7 | THREATS TO VALIDITY

Despite the contributions, this study has several threats to its validity. A possible threat is related to the content validity in this study. We conducted an extensive review of the literature to identify the challenges that negatively affect agile practices for scaling agile methodologies in a distributed environment. This limitation could be mitigated using a systematic literature review.

The construct validity describes whether or not the measurement scale representing the attribute is being measured.<sup>5</sup> The factors highlighted in this study were based on an extensive body of literature<sup>1,11,12,23,27</sup> and were validated through a questionnaire study. The results of the questionnaire study conducted with experts indicated that all the selected factors are related to their work, which we believe is an acceptable level of construct validity.

The external validity explains the generalizability of the research outcomes.<sup>5</sup> In this study, the findings of the literature review were unable to explain the generalizability of the results because of the limited responses obtained from the experts and because most of them were collected from the software industries located in India. However, on the basis of the outcomes of various studies similar in nature to that of our study,<sup>7,41,53</sup> the authors firmly believe that the sample size was sufficiently large to explain the generalizability of this study.

**TABLE 13** Comparison of challenges in proposed framework with studies of Shrivastava and Rathod and chaos report of Standish group

| Proposed Framework of Challenges                         | Shrivastava and Rathod <sup>1</sup> | Standish Group's Chaos Report <sup>52</sup> |
|----------------------------------------------------------|-------------------------------------|---------------------------------------------|
| Lack of management commitments                           | ✓                                   | ✓                                           |
| Lack of effective communication                          | ✓                                   | ✓                                           |
| Lack of knowledge sharing                                | ✓                                   |                                             |
| Cultural differences                                     | ✓                                   |                                             |
| Lack of frequent feedback                                | ✓                                   |                                             |
| Lack of roles and responsibilities                       | ✓                                   |                                             |
| Lack of effective involvement of customer representative | ✓                                   | ✓                                           |
| Inappropriate selection of communication technologies    | ✓                                   |                                             |
| Temporal differences                                     | ✓                                   |                                             |
| Poor requirements and their specifications               | ✓                                   | ✓                                           |
| Lack of trust between clients and vendors                | ✓                                   |                                             |
| Project size and complexity                              | ✓                                   | ✓                                           |
| Improper task allocation                                 | ✓                                   |                                             |
| Inexperienced developers                                 | ✓                                   |                                             |
| Lack of motivational activities                          |                                     |                                             |
| Lack of technological resources                          | ✓                                   | ✓                                           |
| Poor project scope                                       |                                     | ✓                                           |
| Large team size                                          | ✓                                   |                                             |
| Lack of project documentation                            | ✓                                   |                                             |
| Lack of communication training in DSD                    | ✓                                   |                                             |
| Lack of agile process evaluation mechanism               | ✓                                   | ✓                                           |
| Budget constraints                                       |                                     |                                             |

Abbreviation: DSD, distributed software development.

One possible threat is related to the AHP method used for prioritizing the challenges. The prioritization of the challenges was purely based on the discussion and subjective insights developed on the basis of the experts' opinion. Hence, a hasty approach may completely impair the effectiveness of the study, although the CR was calculated for each pairwise comparison matrix, indicating an acceptable internal validity in the priority of the challenges.

## 8 | IMPLICATIONS OF THE STUDY

This work has implications for both researchers and practitioners. The framework presented in this study was based on 22 identified challenges, along with the priorities assigned to them based on the weights calculated using the AHP method. Here, the results of the study provide a knowledge base for both researchers and practitioners in distributed software engineering and agile development. The identification of the challenges for scaling agile methods in a DSD environment is an important contribution for academia. It could assist in the further understanding of the issues/areas that need to be addressed by researchers for scaling agile methods in a DSD environment. Moreover, AHP is a technique used for complex and group decision-making problems; this study can help researches to replicate the AHP method to evaluate the ranking of the challenges based on their importance. The weights and rankings of the challenges and their categories are important from a managerial viewpoint and specific to the case studied, which is a primary contribution that could assist DSD organizations to evaluate and revise their strategies and management approaches to scale agile methods in a DSD environment.

## 9 | CONCLUSIONS AND FUTURE STUDY

Currently, most software development organizations are scaling agile methodologies in a distributed environment because of rapid development rate and low development cost. The increasing trend of using agile methods in a DSD environment motivated us to identify and prioritize the challenges and their categories that could undermine agile practices in a distributed context. The challenges in distributed agile development are the areas or activities that negatively affect agile practices and require management to ensure the successful scaling of agile methodologies in a DSD environment. In this study, we conducted a literature review and questionnaire analysis to explore and validate the challenges. Nevertheless, an AHP method was used to prioritize the challenges using the responses collected from experts based on a pairwise comparison of the challenges. A total

of 22 challenges were identified from the literature review, and these were classified into 4 categories: management, team, technology, and process. The identified challenges were empirically validated using the analysis of 58 questionnaire responses. Finally, the ranking of the challenges was evaluated using the AHP method on the basis of the experts' opinions. The findings of this study indicated that management is the most important category, and the lack of management commitments, lack of effective communication, and temporal differences are the major issues in the management category that affect the scaling programs of agile methodologies in a distributed environment. The weights and rankings of the challenges and their categories provide a framework for distributed organizations to evaluate and revise their management strategies for successfully scaling agile methodologies in a DSD environment. In our methodology, the formulation of the ranking of challenges is completely based on the experts' opinions, and the outcome is based on the views of different experts in the form of a particular rank, which has several benefits, as follows. It is based on a hierarchical and multidimensional model that incorporates the importance of scaling agile methodologies from 4 perspectives: strategic management, characteristics of the team, technology, and the development process. In fact, the given framework emphasizes the importance of success in different categories of challenges to address all of the essential aspects to scale agile methodologies in a DSD environment.

In this study, we explored 22 challenges that negatively influence scaling programs in a DSD environment. The following topics could potentially be part of this study in future work. First, it is necessary to identify the additional challenges and success factors for scaling agile methods in a distributed environment by conducting a systematic literature review and using a large sample size in an empirical study. Second, it is necessary to evaluate the critical challenges, critical success factors, and the best practices for managing them. Third, an agile software process scaling and management model for scaling agile methods in a DSD environment can be developed. This model will be based on the critical challenges, critical success factors, and practices. This model will assist distributed organizations use scaling agile methods in a distributed environment. The findings of the present study can be considered as a first input to develop the agile software process scaling and management model framework.

## ORCID

Mohammad Shameem  <http://orcid.org/0000-0002-6055-5345>

## REFERENCES

- Shrivastava SV, Rathod U. Categorization of risk factors for distributed agile projects. *Inf Softw Technol*. 2015;58:373-387.
- Cockburn A. *Agile Software Development*, Vol. 177. Boston: Addison-Wesley; 2002.
- Dikert K, Paasivaara M, Lassenius C. Challenges and success factors for large-scale agile transformations: a systematic literature review. *J Syst Softw*. 2016;119:87-108.
- Paasivaara M, Durasiewicz S, Lassenius C. Using scrum in a globally distributed project: a case study. *Softw Process Improv Pract*. 2008;13(6):527-544.
- Jalali S, Wohlin C. Global software engineering and agile practices: a systematic review. *J softw Evol Process*. 2012;24(6):643-659.
- Paasivaara M, Behm B, Lassenius C, Hallikainen M. Towards rapid releases in large-scale XaaS development at Ericsson: a case study. In: 2014 IEEE 9th International Conference on Global Software Engineering (ICGSE). Shanghai, China: IEEE; 2014:16-25.
- Niazi M, Mahmood S, Alshayeb M, Riaz MR, Faisal K, Cerpa N, Khan SU, Richardson I. Challenges of project management in global software development: a client-vendor analysis. *Inf Softw Technol*. 2016;80:1-19.
- Herbsleb JD. Global software engineering: the future of socio-technical coordination. In: International Conference on Software Engineering: Future of Software Engineering IEEE Computer Society; 2007; Minneapolis, USA. 188-198.
- Holmstrom H, Conchúir EÓ, Agerfalk J, Fitzgerald B. Global software development challenges: a case study on temporal, geographical and socio-cultural distance. In: International Conference on Global Software Engineering ICGSE'06. Santinho, Florianópolis, Brazil: IEEE; 2006:3-11.
- Khan SU, Niazi M. Critical challenges in offshore software development outsourcing: an empirical study. In: International IASTED Conference on Software Engineering SE; 2012; Greece.
- Shrivastava SV, Rathod U. A risk management framework for distributed agile projects. *Inf Softw Technol*. 2017;85:1-15.
- Thomas G, Botha RA, Greunen D. Understanding the problem of coordination in a large scale distributed environment from a service lens view—towards the South African public sector e-administration criteria for coordination support. *Gov Inf Q*. 2015;32(4):526-538.
- Shrivastava SV, Rathod U. Risks in distributed agile development: a review. *Procedia Soc Behav Sci*. 2014;133:417-424.
- Oliveira Neto FG, Horkoff J, Knauss E, Kasauli R, Liebel G. Challenges of aligning requirements engineering and system testing in large-scale agile: a multiple case study. In: 2017 IEEE 25th International Requirements Engineering Conference Workshops (REW). Lisbon, Portugal: IEEE; 2017:315-322.
- De Brito MF, Da Costa Figueiredo RM, Venson E, Dias Canedo E, Junior MR, Carlos L. Knowledge transfer in a management process for outsourced agile software development. In: Proceedings of the 50th Hawaii International Conference on System Sciences; 2017; Hilton Waikoloa Village, Waikoloa, Hawaii.
- Chau T, Maurer F. Knowledge sharing in agile software teams. *Log Versus Approximation*. 2004;3075:173-183.
- Layman L, Williams L, Damian D, Bures H. Essential communication practices for extreme programming in a global software development team. *Inf Softw Technol*. 2006;48(9):781-794.
- Dybå T, Dingsøyr T. Empirical studies of agile software development: a systematic review. *Inf Softw Technol*. 2008;50(9):833-859.
- Hossain E, Babar MA, Paik H-Y, Verner J. Risk identification and mitigation processes for using scrum in global software development: a conceptual framework. In: Proceedings of the Asia-Pacific Software Engineering Conference, APSEC 2009. Penang, Malaysia: IEEE; 2009:457-464.
- Alzoubi YI, Gill AQ, Al-Ani A. Empirical studies of geographically distributed agile development communication challenges: a systematic review. *Inf Manag*. 2016;53(1):22-37.



21. Laanti M. Agile transformation model for large software development organizations. In: Proceedings of the XP2017 Scientific Workshops. Cologne, Germany: ACM; 2017:19.
22. Ghafoor F, Shah IA, Rashid N. Issues in adopting agile methodologies in global and local software development: a systematic literature review protocol with preliminary results. *Int J Comput Appl*. 2017;160(7).
23. Yagüe A, Garbajosa J, Díaz J, González E. An exploratory study in communication in agile global software development. *Comput Stand Interfaces*. 2016;48:184-197.
24. Paasivaara M, Lassenius C. Scaling scrum in a large globally distributed organization: a case study. In: 2016 IEEE 11th International Conference on Global Software Engineering (ICGSE). Irvine, CA, USA: IEEE; 2016:74-83.
25. Sundararajan S, Bhasi M, Vijayaraghavan PK. Case study on risk management practice in large offshore-outsourced agile software projects. *IET Softw*. 2014;8(6):245-257.
26. Ramesh B, Cao L, Mohan K, Xu P. Can distributed software development be agile? *Commun ACM*. 2006;49(10):41-46.
27. Hoda R, Noble J, Marshall S. The impact of inadequate customer collaboration on self-organizing agile teams. *Inf Softw Technol*. 2011;53(5):521-534.
28. Nerur S, Mahapatra R, Mangalargaj G. Challenges of migrating to agile methodologies. *Commun ACM*. 2005;48(5):72-78.
29. Kasauli R, Liebel G, Knauss E, Gopakumar S, Kanagwa B. Requirements engineering challenges in large-scale agile system development. In: 2017 IEEE 25th International Requirements Engineering Conference (RE). Lisbon, Portugal: IEEE; 2017:352-361.
30. Herbsleb JD, Mockus A. An empirical study of speed and communication in globally distributed software development. *IEEE Trans Softw Eng*. 2003;29(6):481-494.
31. Niinimäki T. Face-to-face, email and instant messaging in distributed agile software development project. In: 2011 sixth IEEE International Conference on Global Software Engineering Workshop (ICGSEW). IEEE; 2011:78-84. <https://www.computer.org/>.
32. Saaty TL. *What is the analytic hierarchy process?* Mathematical models for decision support. Berlin Heidelberg: Springer; 1988:109-121.
33. Kabra G, Ramesh A, Arshinder K. Identification and prioritization of coordination barriers in humanitarian supply chain management. *Int J Disaster Risk Reduct*. 2015;13:128-138.
34. Albayrak E, Erensal YC. Using analytic hierarchy process (AHP) to improve human performance: an application of multiple criteria decision making problem. *J Intell Manuf*. 2004;15(4):491-503.
35. Chan FT, Kumar N. Global supplier development considering risk factors using fuzzy extended AHP-based approach. *Omega*. 2007;35(4):417-431.
36. Bozburu FT, Beskese A, Kahraman C. Prioritization of human capital measurement indicators using fuzzy AHP. *Expert Syst Appl*. 2007;32(4):1100-1112.
37. Badri MA. A combined AHP-GP model for quality control systems. *Int J Prod Econ*. 2001;72(1):27-40.
38. Khan AA, Keung J, Niazi M, Hussain S, Ahmad A. Systematic literature review and empirical investigation of barriers to process improvement in global software development: client-vendor perspective. *Inf Softw Technol*. 2017;87:180-205.
39. Ghazi AN, Petersen K, Reddy SSVR, Nekkanti H. Survey research in software engineering: problems and strategies. *arXiv preprint arXiv:1704.01090*. 2017. Preprint submitted to e-Informatica Software Engineering Journal, February 201
40. Khan AA, Keung JW, Abdullah-Al-Wadud M. SPIIMM: toward a model for software process improvement implementation and management in global software development. *IEEE Access*. 2017;5:13720-13741.
41. Khan SU, Niazi M, Ahmad R. Empirical investigation of success factors for offshore software development outsourcing vendors. *IET Softw*. 2012;6(1):1-15.
42. Kontio J, Hoglund M, Ryden J, Abrahamsson P. Managing commitments and risks: challenges in distributed agile development. In: Proceedings 26th International Conference on Software Engineering, ICSE 2004. Edinburgh, UK, UK: IEEE; 2004:732-733.
43. Chow T, Cao D-B. A survey study of critical success factors in agile software projects. *J Syst Softw*. 2008;81(6):961-971.
44. Kuusinen K, Mikkonen T, Pakarinen S. *Agile User Experience Development in a Large Software Organization: Good Expertise but Limited Impact*. Toulouse: Human-Centered Software Engineering; 2012:94-111.
45. Begel A, Nagappan N. Global software development: who does it? In: IEEE International Conference on Global Software Engineering, ICGSE 2008. Bangalore, India: IEEE; 2008:195-199.
46. Shameem M, Kumar C, Chandra B. Communication related issues in GSD: an exploratory study. In: 2015 9th International Conference on Software, Knowledge, Information Management and Applications (SKIMA). Kathmandu, Nepal: IEEE; 2015:1-5.
47. Mishra D, Mishra A, Ostrovska S. Impact of physical ambiance on communication, collaboration and coordination in agile software development: an empirical evaluation. *Inf Softw Technol*. 2012;54(10):1067-1078.
48. Musio I. IBM industry practice: challenges in offshore software development from a global delivery center. *Seafood*: Springer; 2009;Zurich, Switzerland:4-13.
49. Ågerfalk PJ, Fitzgerald B, Slaughter SA. Introduction to the special issue—a flexible and distributed information systems development: state of the art and research challenges. *Inf Syst Res*. 2009;20(3):317-328.
50. Boehm B. Get ready for agile methods, with care. *Comput*. 2002;35(1):64-69.
51. Bayona-Oré S, Calvo-Manzano JA, Cuevas G, San-Feliu T. Critical success factors taxonomy for software process deployment. *Softw Qual J*. 2014;22(1):21-48.
52. Hastie S, Wojewoda S. Standish group 2015 chaos report—Q&A with Jennifer Lynch. Retrieved. 2015;1(15):2016.
53. Yaghoobi T. Prioritizing key success factors of software projects using fuzzy AHP. *J Softw Evol Process*. 2017;30(1):e1891.

**How to cite this article:** Shameem M, Kumar RR, Kumar C, Chandra B, Khan AA. Prioritizing challenges of agile process in distributed software development environment using analytic hierarchy process. *J Softw Evol Proc*. 2018;30:e1979. <https://doi.org/10.1002/smr.1979>

**SECTION 1: DEMOGRAPHICS OF THE RESPONDENTS**

Name (optional)

Please specify your job title.

What is your experience (in years) of working in distributed agile project?

Please specify your country.

|                                                                                          |                             |                      |                                |
|------------------------------------------------------------------------------------------|-----------------------------|----------------------|--------------------------------|
| Please specify the primary business functions of your company.                           | Global software outsourcing | Offshore development | In-house development           |
| Please specify the size of your organization.                                            | Large                       | Medium               | Small                          |
| Approximately how many staff are employed by your company? (Please tick as appropriate.) | Less than 30                | 30-50                | 51 - 100                       |
|                                                                                          | 101-150                     | 151-200              | More than 200                  |
| Which agile method is being followed in project?                                         | Extreme programming (XP)    | Scrum                | Other, please specify the name |

**CHALLENGES IN THE DISTRIBUTED AGILE DEVELOPMENT**

In the following sections, various challenges, along with their categories, which may affect the agile process while applying in DSD environment, have been presented.

**SECTION 2: MANAGEMENT-RELATED CHALLENGES**

In this section, the challenges related to the "management" category are given below.

Please cross the appropriate box based on your experience in distributed agile development.

SD: strongly disagree; D: disagree; N: neutral; A: agree; SA: strongly agree

| Challenges                      | SD | D | N | A | SA |
|---------------------------------|----|---|---|---|----|
| Cultural differences            |    |   |   |   |    |
| Lack of motivational activities |    |   |   |   |    |
| Lack of management commitments  |    |   |   |   |    |
| Lack of effective communication |    |   |   |   |    |
| Temporal differences            |    |   |   |   |    |

**SECTION 3: TEAM-RELATED CHALLENGES**

In this section, the challenges related to the "team" category are given below.

Please cross the appropriate box based on your experience in distributed agile software development.

SD: strongly disagree; D: disagree; N: neutral; A: agree; SA: strongly agree

| Challenges                                               | SD | D | N | A | SA |
|----------------------------------------------------------|----|---|---|---|----|
| Lack of knowledge sharing                                |    |   |   |   |    |
| Large team size                                          |    |   |   |   |    |
| Lack of effective involvement of customer representative |    |   |   |   |    |
| Inexperienced developers                                 |    |   |   |   |    |
| Improper task allocation                                 |    |   |   |   |    |
| Lack of frequent feedback                                |    |   |   |   |    |
| Lack of trust between customers and vendors              |    |   |   |   |    |

**SECTION 4: TECHNOLOGY-RELATED CHALLENGES**

In this section, the challenges related to the "technology" category are given below.

Please cross the appropriate box based on your experience in a distributed agile software development.

SD: strongly disagree; D: disagree; N: neutral; A: agree; SA: strongly agree

| Challenges                                            | SD | D | N | A | SA |
|-------------------------------------------------------|----|---|---|---|----|
| Lack of communication training in DSD                 |    |   |   |   |    |
| Lack of technological resources                       |    |   |   |   |    |
| Budget constraints                                    |    |   |   |   |    |
| Inappropriate selection of communication technologies |    |   |   |   |    |

**SECTION 5: PROCESS-RELATED CHALLENGES**

In this section, the challenges related to the "process" category are given below.

Please cross the appropriate box based on your experience in distributed agile software development.

SD: strongly agree; D: disagree; N: neutral; A: agree; SA: strongly agree

| Challenges                                                                           | SD | D | N | A | SA |
|--------------------------------------------------------------------------------------|----|---|---|---|----|
| Poor project scope                                                                   |    |   |   |   |    |
| Lack of project documentation                                                        |    |   |   |   |    |
| Lack of agile process evaluation mechanism                                           |    |   |   |   |    |
| Poor requirements and their specifications                                           |    |   |   |   |    |
| Lack of roles and responsibilities                                                   |    |   |   |   |    |
| Project size and complexity                                                          |    |   |   |   |    |
| Other, if any, that is not mentioned in any of the above categories (please specify) |    |   |   |   |    |

## APPENDIX B: INFORMATION OF RESPONDENTS

| S. No. | Job Title           | Working Experience, y | Country    | Size of the Company | Primary Business Functions of the Company | Agile Method             |
|--------|---------------------|-----------------------|------------|---------------------|-------------------------------------------|--------------------------|
| 1      | Software engineer   | 6                     | India      | Large               | Global outsourcing                        | Scrum                    |
| 2      | Analytic consultant | 5                     | USA        | Large               | Global outsourcing /collocated            | Scrum                    |
| 3      | IT engineer         | 4                     | Netherland | Medium              | Global outsourcing                        | Extreme programming (XP) |
| 4      | Software engineer   | 2                     | India      | Medium              | Global outsourcing /collocated            | Extreme programming (XP) |
| 5      | Assistant professor | 4                     | Turkey     | Large               | Global outsourcing /collocated            | Scrum                    |
| 6      | Software engineer   | 5                     | India      | Medium              | Global outsourcing /collocated            | Extreme programming (XP) |
| 7      | Lecturer            | 5                     | Malaysia   | Medium              | Global outsourcing                        | Extreme programming (XP) |
| 8      | IT coordinator      | 12                    | Australia  | Medium              | Global outsourcing /collocated            | Scrum                    |
| 9      | Team member         | 3                     | India      | Medium              | Global outsourcing /collocated            | Extreme programming (XP) |
| 10     | Assistant professor | 10                    | Pakistan   | Large               | Global outsourcing /collocated            | Extreme programming (XP) |
| 11     | Professor           | 12                    | India      | Large               | Global outsourcing                        |                          |
| 12     | Associate professor | 4                     | Malaysia   | Large               | Global outsourcing                        | Scrum                    |
| 13     | Software engineer   | 7                     | Pakistan   | Small               | Collocated                                | Extreme programming (XP) |
| 14     | Asst professor      | 4                     | India      | Large               | Global outsourcing /collocated            | Scrum                    |
| 15     | Researcher          | 2                     | Hong Kong  | Large               | Global outsourcing /collocated            | Scrum                    |
| 16     | Software engineer   | 13                    | China      | Large               | Global outsourcing /collocated            | Extreme programming (XP) |
| 17     | Team lead           | 4                     | India      | Medium              | Global outsourcing                        | Scrum                    |
| 18     | Manager             | 6                     | China      | Large               | Global outsourcing /collocated            | Lean                     |
| 19     | Team lead           | 4                     | India      | Large               | Global outsourcing /collocated            | Other                    |
| 20     | Tester              | 3                     | UAE        | Medium              | Global outsourcing                        | Scrum                    |
| 21     | Senior lecturer     | 4                     | UAE        | Medium              | Global outsourcing                        | Extreme programming (XP) |
| 22     | Software developer  | 7                     | Pakistan   | Small               | Collocated                                | Scrum                    |
| 23     | Team lead           | 5                     | India      | Medium              | Global outsourcing                        | Other                    |
| 24     | Manager             | 6                     | Australia  | Large               | Global outsourcing                        | Scrum                    |
| 25     | Scrum master        | 4                     | Netherland | Large               | Global outsourcing /collocated            | Extreme programming (XP) |
| 26     | Web developer       | 5                     | India      | Large               | Global outsourcing                        | Scrum                    |
| 27     | Research student    | 2                     | China      | Large               | Global outsourcing /collocated            | Scrum                    |
| 28     | Academician         | 4                     | Pakistan   | Medium              | Collocated                                | Lean                     |
| 29     | Software engineer   | 5                     | UAE        | Large               | Global outsourcing                        | Extreme programming (XP) |
| 30     | Software developer  | 2                     | Sydney     | Medium              | Global outsourcing /collocated            | Scrum                    |

| S. No. | Job Title                  | Working Experience, y | Country    | Size of the Company | Primary Business Functions of the Company | Agile Method             |
|--------|----------------------------|-----------------------|------------|---------------------|-------------------------------------------|--------------------------|
| 31     | Team lead                  | 5                     | India      | Small               | Global outsourcing /collocated            | Extreme programming (XP) |
| 32     | Manager                    | 4                     | India      | Large               | Global outsourcing /collocated            | Scrum                    |
| 33     | Scrum master               | 8                     | Australia  | Medium              | Global outsourcing /collocated            | Extreme programming (XP) |
| 34     | Web developer              | 4                     | India      | Small               | Global outsourcing /collocated            | Scrum                    |
| 35     | Mobile developer           | 5                     | India      | Medium              | Global outsourcing /collocated            | Scrum                    |
| 36     | Senior developer           | 5                     | UAE        | Large               | Global outsourcing                        | Other                    |
| 37     | Team member                | 18                    | Scotland   | Large               | Global outsourcing /collocated            | Lean                     |
| 38     | Web developer              | 6                     | India      | Large               | Global outsourcing                        | Extreme programming (XP) |
| 39     | Senior manager             | 3                     | Australia  | Large               | Global outsourcing /collocated            | Lean                     |
| 40     | iOS developer              | 3                     | Netherland | Medium              | Global outsourcing /collocated            | Extreme programming (XP) |
| 41     | Android developer          | 4                     | India      | Large               | Global outsourcing                        | Scrum                    |
| 42     | Chairman and MD            | 7                     | China      | Large               | Global outsourcing /collocated            | Extreme programming (XP) |
| 43     | Senior software developer  | 5                     | Pakistan   | Medium              | Global outsourcing                        | Extreme programming (XP) |
| 44     | Graphic designer           | 6                     | UAE        | Medium              | Global outsourcing /collocated            | Scrum                    |
| 45     | Manager                    | 4                     | Sydney     | Small               | Collocated                                | Extreme programming (XP) |
| 46     | HOD                        | 10                    | India      | Medium              | Global outsourcing /collocated            | Other                    |
| 47     | Graphic designer           | 3                     | India      | Large               | Global outsourcing                        | Extreme programming (XP) |
| 48     | SAP consultant             | 4                     | Australia  | Large               | Global outsourcing /collocated            | Scrum                    |
| 49     | MD of Artzzasia handicraft | 3                     | India      | Large               | Global outsourcing                        | Other                    |
| 50     | Agile consultant           | 5                     | Australia  | Large               | Global outsourcing /collocated            | Scrum                    |
| 51     | Data base administrator    | 6                     | Netherland | Medium              | Global outsourcing /collocated            | Extreme programming (XP) |
| 52     | Software engineer          | 3                     | India      | Large               | Global outsourcing                        | Lean                     |
| 53     | Project manger             | 8                     | China      | Large               | Global outsourcing /collocated            | Scrum                    |
| 54     | Scrum Master               | 3                     | Pakistan   | Medium              | Global outsourcing /collocated            | Extreme programming (XP) |
| 55     | Asst manager               | 3                     | UAE        | Medium              | Global outsourcing /collocated            | Lean                     |
| 56     | Software engineer          | 4                     | Sydney     | Small               | Collocated                                | Scrum                    |
| 57     | Software developer         | 9                     | India      | Medium              | Global outsourcing /collocated            | Extreme programming (XP) |
| 58     | Software developer         | 6                     | India      | Large               | Global outsourcing /collocated            |                          |